

# Cupcast II: Validation and Calibration

Carlos Garcia

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## Abstract

I validate the forecasting model of Paper I in two tiers. Tier 1 tests the match-level engine where data is abundant: club football (league seasons and continental cups), scored against bookmaker closing odds. Tier 2 replays Euro 2024 and Copa América 2024 end-to-end with training windows truncated to each tournament's eve, testing the full international pipeline including tournament simulation. Probabilistic skill is measured with strictly proper scoring rules [1] — log-loss and Brier score — alongside reliability diagrams; miscalibration, if found, is corrected by isotonic regression [3] or Platt scaling [2] before any 2026 forecast is issued.

## 1 Introduction

*To be written.*

## 2 Validation Design

*To be written.*

## 3 Scoring Rules

*To be written.*

## 4 Tier 1: Club Football

*To be written.*

## 5 Tier 2: Tournament Replays

*To be written.*

## 6 Calibration

*To be written.*

## 7 Discussion

*To be written.*

## References

- [1] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [2] J. C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.
- [3] B. Zadrozny and C. Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002. doi: 10.1145/775047.775151.